EDGE AI-BASED REAL-TIME ARM MOVEMENT QUALITY ASSESSMENT SYSTEM FOR UPPER LIMB REHABILITATION

A Research Internship Report

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Declaration of Originality

I hereby declare that this research internship report titled "Edge AI-Based Real-Time Arm Movement Quality Assessment System for Upper Limb Rehabilitation" is my own original work carried out during my internship at Digital Pakistan Lab (DPL), Rawalpindi, Pakistan, from June to August 2026. This report has not been submitted elsewhere for any academic award or qualification.

All sources of information, including published papers, articles, and technical documentation, have been duly acknowledged through proper citations in accordance with academic convention.

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Abstract

Upper limb rehabilitation requires patients to perform repetitive arm exercises with precise movement quality, yet conventional therapy relies entirely on therapist supervision and provides no objective, quantitative feedback on movement correctness. This report presents the design, development, and evaluation of an Edge AI-based wearable system for real-time arm movement quality assessment, implemented on the Arduino Nano 33 BLE Rev2 microcontroller. The system employs the onboard BMI270 Inertial Measurement Unit (IMU) to capture six-axis motion data at 59 Hz, which is processed by a trained neural network deployed directly on the device using the Edge Impulse platform.

A novel hybrid classification architecture is proposed that combines physics-based threshold detection for arm positional state with a machine learning model for movement quality assessment. The quality model classifies arm raises into five categories: good raise, fast raise, jerky raise, incomplete raise, and full extension hold. A total of 230 labelled samples across five quality classes were collected from three subjects. The trained model achieves a validation accuracy of 98.6% with a loss of 0.04 with an area under the ROC curve of 1.00, while operating within the severe resource constraints of the target microcontroller — requiring only 1.7 KB of RAM and 18.9 KB of flash memory, with an inference latency of 1 millisecond.

A key research contribution of this work is the identification that machine learning alone is insufficient for detecting incomplete raises — a clinically critical condition — due to identical motion signatures between complete and incomplete raises. A physics-based endpoint detection mechanism is introduced to override model classification using accelerometer gravity sensing, achieving robust detection of incomplete extension. The system outputs a per-repetition quality score and real-time textual feedback, enabling unsupervised home-based rehabilitation exercise. Future work includes integration of a force-sensitive resistor (FSR) for grip strength measurement, multi-patient calibration, and BLE connectivity to a mobile application for therapist-side session review.

Keywords: Edge AI, TinyML, upper limb rehabilitation, gesture recognition, IMU, Arduino, movement quality assessment, wearable sensing, Edge Impulse, hybrid classification.

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1. Introduction

1.1 Background and Motivation

Upper limb impairment is among the most common and debilitating consequences of neurological and musculoskeletal conditions, including stroke, cerebral palsy, multiple sclerosis, and rotator cuff injuries. Globally, stroke alone affects over 15 million people annually, with approximately two-thirds experiencing some degree of upper limb motor dysfunction that impairs their ability to perform activities of daily living [1]. Effective rehabilitation requires patients to perform high-repetition exercises under controlled conditions — studies suggest that neuroplasticity-driven motor recovery requires hundreds to thousands of movement repetitions per day, far exceeding what is feasible in supervised clinical sessions [2].

Conventional rehabilitation practice is constrained by the availability of trained physiotherapists, geographical access to clinical facilities, and the prohibitive cost of extended therapy programmes. Patients performing exercises at home receive no objective feedback on whether their movements are executed correctly, leading to the reinforcement of compensatory movement patterns that can retard recovery or cause secondary injury. There is therefore a compelling clinical and technical need for low-cost, intelligent wearable systems capable of monitoring movement quality in real time, without requiring continuous therapist supervision.

1.2 Problem Statement

Existing home-based rehabilitation technologies fall into two broad categories: simple repetition counters that detect only whether a movement occurred, and expensive clinical-grade motion capture systems that require controlled environments and expert operators. Neither category addresses the core clinical requirement: real-time, objective assessment of movement quality during unsupervised exercise. A patient who completes a repetition too rapidly, with jerky uncontrolled motion, or without achieving full range of motion receives no different feedback from a patient who performs the movement correctly — yet the therapeutic benefit of these movements differs substantially.

Furthermore, deploying artificial intelligence for this purpose presents unique engineering constraints. Healthcare wearables must operate without network connectivity to protect patient data privacy, must function under embedded resource limitations of microcontrollers, and must deliver near-instantaneous feedback to be therapeutically useful. These requirements point toward Edge AI — the deployment of machine learning models directly on resource-constrained edge devices — as the appropriate technical paradigm.

1.3 Research Objectives

This research addresses the following primary objectives:

- Design and implement a wearable IMU-based motion sensing system on the Arduino Nano 33 BLE Rev2 for real-time arm movement data acquisition.
- Develop a machine learning pipeline on the Edge Impulse platform to classify arm movement quality into clinically meaningful categories.
- Investigate the limitations of pure model-based classification for clinically critical movement states and propose a hybrid physics-AI architecture as a solution.

- Evaluate system performance in terms of classification accuracy, inference latency, and on-device resource utilisation.
- Demonstrate the feasibility of a complete, self-contained rehabilitation feedback system operating within the computational constraints of a microcontroller.

1.4 Scope and Delimitations

This work focuses on a single upper limb exercise — a controlled arm raise from a resting position at the side of the body to full horizontal extension — which forms a foundational component of shoulder and rotator cuff rehabilitation protocols. The system is designed for deployment on a single wrist-mounted device. Grip strength measurement via force-sensitive resistor, which forms a planned extension of this work, falls outside the current scope. The system was evaluated on a small cohort of three healthy subjects and has not undergone clinical validation with patient populations; this is identified as a priority direction for future work.

1.5 Report Structure

The remainder of this report is organised as follows. Section 2 reviews related work in wearable rehabilitation sensing, IMU-based gesture recognition, and Edge AI for healthcare. Section 3 presents the overall system architecture and hardware design. Section 4 describes the data collection methodology, machine learning pipeline, and hybrid classification architecture. Section 5 presents experimental results and performance analysis. Section 6 discusses findings, limitations, and clinical implications. Sections 7 and 8 provide the conclusion and future work directions respectively.

2. Literature Review

2.1 Wearable Systems for Upper Limb Rehabilitation

The use of wearable sensors for upper limb rehabilitation monitoring has attracted substantial research attention over the past decade. A comprehensive systematic review by Milosevic et al. (2024) covering research from 2011 to 2023 identified IMU-based wrist and arm sensing as the dominant modality for motion tracking in rehabilitation contexts, noting that real-time feedback capability is a key feature distinguishing clinically effective systems from passive monitoring solutions [3]. The review emphasised that accuracy of system measurements is a prerequisite for therapeutic utility, as inaccurate feedback may reinforce incorrect movement patterns.

Kanzler et al. (2024) demonstrated the feasibility of classifying upper limb movement quality at home using a single wrist-worn IMU in a cohort of nineteen stroke survivors, showing that a wearable IMU can distinguish compliant from non-compliant movement execution compared to in-clinic reference patterns [4]. This work is particularly relevant to the present study as it validates the clinical premise that a single low-cost IMU sensor carries sufficient information to assess rehabilitation movement quality without multichannel sensor arrays. Their findings motivate the single-board approach adopted in this work, while their reliance on Dynamic Time Warping (DTW) for comparison against reference templates — a computationally intensive operation — highlights the advantage of the neural network approach for real-time embedded deployment.

Ferreira et al. (2023) proposed an affordable FSR-based wearable feedback system for upper extremity rehabilitation, demonstrating real-time classification of six postures relevant to a functional drinking task using forearm force myography [5]. While the sensor modality differs from IMU-based inertial sensing, their work establishes the clinical value of posture classification for exercise quality assessment and motivates the inclusion of force sensing as a future enhancement in the present system.

2.2 IMU-Based Gesture and Motion Recognition

Inertial Measurement Units combining accelerometer and gyroscope measurements have been extensively studied for gesture and activity recognition in human-machine interaction contexts. A comprehensive review by Zhang et al. (2024) on sEMG-IMU sensor fusion for upper limb pattern recognition highlighted that IMU-only systems — while carrying less physiological information than electromyography — offer significantly lower cost, simpler placement, and higher subject comfort, making them appropriate for home-based and long-duration monitoring applications [6].

Chang et al. (2024) demonstrated real-time arm motion tracking and hand gesture recognition using a single IMU, confirming that six-axis inertial data (three-axis accelerometer combined with three-axis gyroscope) provides sufficient signal richness for reliable gesture classification across multiple subjects [7]. The critical contribution of gyroscope data to motion quality discrimination — particularly the ability to distinguish controlled from rapid or interrupted movements through analysis of angular velocity profiles — is a finding directly exploited in the present work.

Occhipinti et al. (2024) proposed an efficient IMU-based hand gesture recognition system using deep learning, reporting high accuracy on a microcontroller target device and demonstrating the viability of neural network inference at the embedded edge for gesture

classification tasks [8]. Their work provided important precedent for the architecture design choices made in the present study.

2.3 Edge AI and TinyML for Healthcare

The emergence of TinyML — the deployment of machine learning models on microcontroller-class hardware — has opened new possibilities for intelligent healthcare sensing that operates without cloud connectivity. Abadade et al. (2023) provided a comprehensive survey of TinyML approaches, identifying healthcare monitoring as one of the most impactful application domains and noting that model quantisation — the conversion of floating-point weights to integer representations — is essential for fitting inference pipelines within the memory constraints of microcontrollers such as the ARM Cortex-M4 series [9].

The integration of Edge AI with wearable health devices was comprehensively reviewed by Peng et al. (2025), who demonstrated that deploying AI models at the edge rather than in the cloud offers three critical advantages for rehabilitation applications: elimination of network latency (enabling sub-100ms feedback loops), preservation of patient data privacy (no transmission of sensitive health data), and continuous operation in environments without internet connectivity [10]. These advantages are directly realised in the present system, which performs all inference on-device with a measured latency of 1 millisecond.

Jiang et al. (2021) demonstrated a wearable on-device deep learning system for hand gesture recognition using a low-power processor, explicitly noting that conventional cloud-based approaches are "not convenient for remote follow-up treatment of movement rehabilitation training" and motivating the on-device paradigm [11]. The Edge Impulse platform, used in the present work, provides an end-to-end pipeline for data collection, model training, and deployment that has been identified in a thematic analysis of TinyML research as a leading tool for embedded machine learning on Arduino-class hardware [12].

Lee et al. (2024) demonstrated a deep learning model for classifying shoulder pain rehabilitation exercises using IMU sensors, achieving high classification accuracy for distinguishing correct from incorrect exercise execution [13]. This work is closely aligned with the objectives of the present study and confirms the viability of neural network approaches for exercise quality classification from wearable IMU data.

2.4 Automatic Rehabilitation Exercise Assessment

Wu et al. (2024) proposed a lightweight multichannel 1D-CNN framework for automatic rehabilitation exercise task assessment of stroke patients using wearable sensors, demonstrating that action quality assessment — going beyond simple repetition counting to evaluate the correctness of movement execution — is achievable with low-cost wearable hardware [14]. Their system addresses the same fundamental gap that motivates the present work: the absence of objective, automated quality assessment in home-based rehabilitation contexts.

A recurring limitation identified across the reviewed literature is the tendency for systems to be evaluated on small, homogeneous subject populations, raising questions about generalisation to the broader clinical population. The present work acknowledges this limitation and addresses it in part through multi-subject data collection, while identifying expanded clinical validation as a priority for future development.

2.5 Research Gap

The reviewed literature, while demonstrating strong technical foundations for IMU-based gesture recognition and Edge AI deployment, reveals an underexplored area: the combination of physics-based signal interpretation with learned neural network classification in a hybrid architecture specifically designed to handle the ambiguity of clinically critical movement states such as incomplete range of motion. Existing systems either rely purely on model output — which, as demonstrated in this work, fails for states with identical motion signatures — or use only physics-based thresholds, which cannot capture the nuanced quality distinctions that drive therapeutic outcomes. The present work directly addresses this gap through the proposed hybrid physics-AI classification architecture.

3. System Design and Architecture

3.1 Overall System Architecture

The proposed system follows a layered Edge AI architecture in which all sensing, feature extraction, machine learning inference, and feedback generation occur on-device without any network dependency. Figure 1 illustrates the high-level data flow.

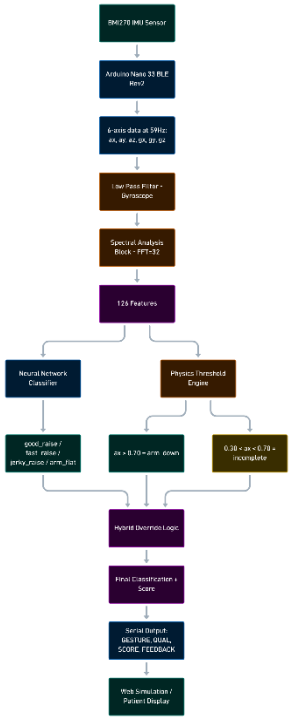


Figure 1: Edge AI Rehabilitation System Architecture Block Diagram

The system operates through four functional layers:

- Sensing Layer: The BMI270 IMU captures six-axis motion data (three-axis accelerometer, three-axis gyroscope) at 59 Hz over the I2C bus on the Arduino Nano 33 BLE Rev2.
- Signal Processing Layer: Raw sensor data is conditioned through a first-order low-pass filter applied to gyroscope channels, followed by windowed spectral feature extraction as configured in the Edge Impulse impulse design.
- Intelligence Layer: A dual-path classification system comprising a trained neural network for movement quality assessment and a physics-based threshold engine for arm position and endpoint detection operates in parallel on each 3-second observation window.
- Output Layer: Classification results, quality scores, and patient feedback messages are transmitted over the UART serial interface at 115,200 baud to a connected visualisation interface.

3.2 Hardware Platform

The Arduino Nano 33 BLE Rev2 was selected as the target platform on the basis of its integration of an advanced IMU sensor, Bluetooth Low Energy radio, and ARM Cortex-M4 processor within a compact form factor suitable for wrist-worn deployment.

Component	Specification
Processor	nRF52840, ARM Cortex-M4 @ 64 MHz
RAM	256 KB SRAM
Flash	1 MB
IMU	BMI270 (accelerometer + gyroscope) + BMM150 (magnetometer)
Accelerometer range	±2g / ±4g / ±8g / ±16g (configurable)
Gyroscope range	±125 / ±250 / ±500 / ±1000 / ±2000 dps
Connectivity	Bluetooth Low Energy 5.0
Interface	USB via UART bridge, I2C, SPI, UART
Dimensions	45 mm × 18 mm
Operating voltage	3.3V

Table 1: Arduino Nano 33 BLE Rev2 Hardware Specifications

3.3 Sensor Configuration and Mounting

The BMI270 IMU was configured to output all six axes — accelerometer (ax, ay, az) and gyroscope (gx, gy, gz) — at a nominal sampling frequency of 59 Hz, corresponding to the closest achievable rate to the target 60 Hz given the BMI270's fixed output data rate options. The board was mounted on the dorsal surface of the dominant hand using a firm elastic strap, with the USB port oriented toward the fingers. This mounting orientation was selected to maximize the discriminability of the accelerometer ax axis for arm position detection, as measurements confirmed ax ≈ +0.98 to +1.0g when the arm is fully extended downward (gravity aligned with the long axis of the board) and ax ≈ 0.00 to +0.25g when the arm is raised to the horizontal position.

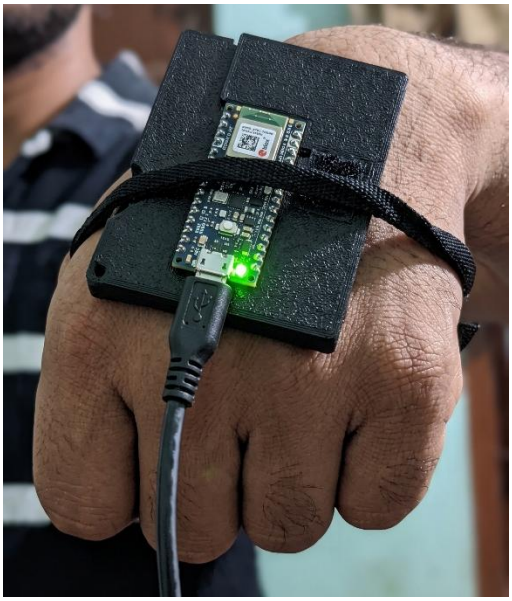


Figure 2: Arduino Nano 33 BLE Rev2 mounted on dorsal surface of hand with elastic strap, USB port oriented toward fingers

Board mounting stability is critical for gyroscope signal quality. Pilot experiments revealed that loose mounting on a compliant surface (such as foam) introduces high-frequency vibrational artefacts in gyroscope channels that degrade classification accuracy. A firm strap that prevents independent board movement relative to the hand was found to substantially reduce this noise, and a first-order digital low-pass filter with smoothing coefficient $\alpha = 0.2$ was applied to all gyroscope channels to further attenuate residual vibration artefacts while preserving the rotational motion signatures relevant to gesture classification.

3.4 Hybrid Classification Architecture

A central design contribution of this work is the hybrid physics-AI classification architecture, which arose from the empirical observation that a machine learning model trained on motion features alone cannot reliably distinguish a complete arm raise that stops prematurely (incomplete raise) from a smooth complete raise, because the motion signatures of both are identical — the difference lies in the final arm position, not in the movement pattern. This insight motivated a two-path classification strategy:

- Path 1 — Neural Network: Classifies movement quality (good, fast, jerky) from the spectral features of the 3-second observation window. This path has strong discriminative power for motion pattern differences.
- Path 2 — Physics Engine: Determines arm endpoint position from the accelerometer ax reading averaged over 10 samples at the end of the window. This path has perfect discriminative power for the incomplete raise condition, which is invisible to Path 1.

The outputs of both paths are combined through a rule-based override logic: if the physics engine detects the arm endpoint in the halfway zone ($0.30 < ax < 0.70g$), the classification is overridden to incomplete raise regardless of the neural network output. This hybrid approach exploits the complementary strengths of data-driven and physics-based reasoning, representing a principled solution to a fundamental limitation of pure machine learning for this application.

4. Methodology

4.1 Data Acquisition Protocol

Data collection was conducted using the Edge Impulse Data Forwarder tool, which reads the six-axis CSV output from the Arduino serial port and streams it directly into the Edge Impulse project dashboard for labelling and storage. The Arduino firmware was configured to print all six axis values at 59 Hz in the format: ax, ay, az, gx, gy, gz.

All data collection was performed with the board mounted on the dorsal surface of the hand using an elastic strap, in a seated position to reflect the expected clinical deployment context. Three subjects participated in data collection: the primary investigator and two additional volunteers (one male, one female), introducing inter-subject variability in movement speed, arm length, and natural movement patterns. No specific screening criteria were applied beyond the ability to perform the target exercise without pain.

4.2 Quality Model — Class Definitions and Collection Protocol

Five movement quality classes were defined based on clinically relevant movement deviations identified in consultation with rehabilitation literature:

Class	Definition	Clinical Relevance	Samples
good_raise	Smooth, controlled arm raise from vertical to horizontal, 2–3 seconds duration	Target movement — full therapeutic benefit	50
fast_raise	Arm raised to horizontal in under 1 second (rapid snap)	Risk of rotator cuff strain; reduced motor control benefit	45
jerky_raise	Arm raise with 2–3 deliberate stops and restarts mid-movement	Indicates muscle fatigue, pain avoidance, or poor motor control	45
incomplete_raise	Arm raise that stops at approximately 45° and holds, never reaching horizontal	Insufficient range of motion; common in stiff or painful shoulder	45
arm_flat	Arm held steady at full horizontal extension for 2–3 seconds	Isometric hold component of rehabilitation protocol	45

Table 2: Quality Model Class Definitions and Sample Counts

Each sample was recorded at 2–3 seconds duration, beginning at the moment of movement initiation and terminating at the point of completion or maximum reach. Samples were recorded in batches of 10, with rest periods between batches to avoid fatigue-induced movement pattern drift. Samples exhibiting artefacts from board readjustment or accidental contact were visually inspected in the Edge Impulse raw data viewer and deleted prior to training. A total of 230 samples were collected across all five classes.

4.3 Edge Impulse Machine Learning Pipeline

4.3.1 Impulse Design

The Edge Impulse impulse was configured with the following parameters:

Parameter	Value	Rationale
Window size	3000 ms	Covers complete gesture including slow raises
Window increase (stride)	200 ms	Overlapping windows for robust continuous classification
Sampling frequency	59 Hz	Matches data collection frequency
Input axes	ax, ay, az, gx, gy, gz	All six axes required for quality discrimination
Train/test split	80% / 20%	Standard supervised learning partition

Table 3: Edge Impulse Impulse Configuration

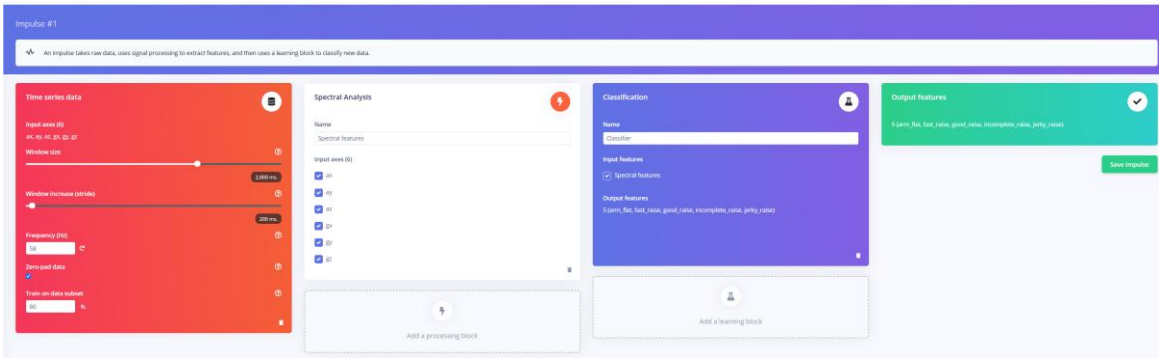


Figure 3: Edge Impulse Impulse Design — Time Series Input → Spectral Analysis → Classification with 6-axis input configuration

4.3.2 Signal Processing Block — Spectral Analysis

A Spectral Analysis processing block was selected over a raw data block for the following reasons. The spectral approach converts the time-domain sensor signal into frequency-domain features via Fast Fourier Transform (FFT), which captures the characteristic frequency content of each movement quality class. A fast raise produces a single high-amplitude, narrow-bandwidth spectral peak in the gyroscope channels, reflecting a brief intense rotational event. A jerky raise produces multiple lower-amplitude peaks distributed across the window. A good raise produces a broad, moderate-amplitude spectral distribution. These frequency-domain representations are substantially more discriminative than raw time-series values for the classification task.

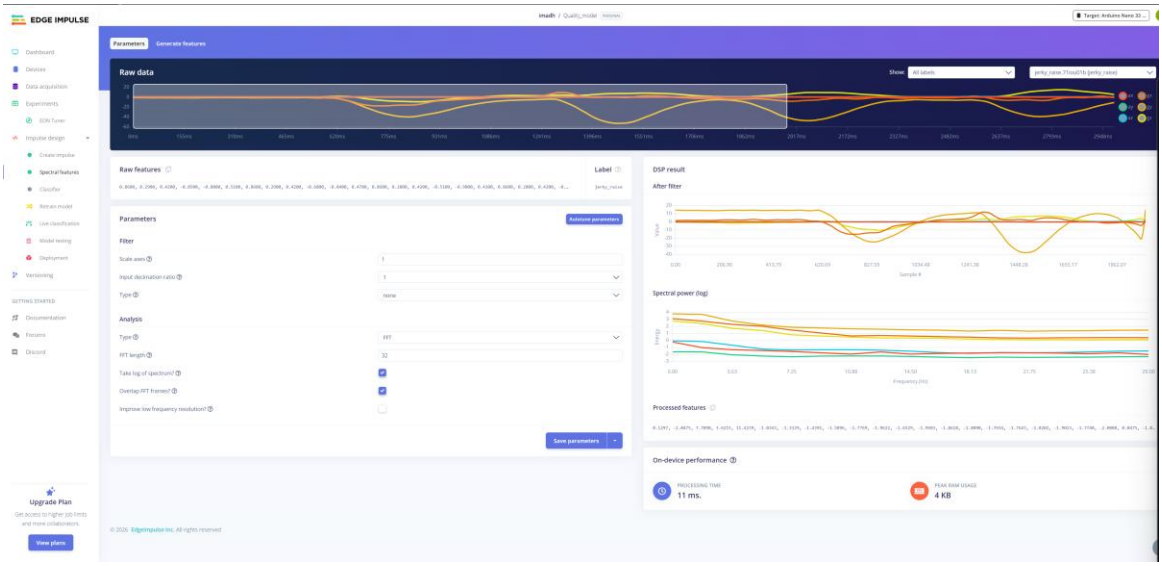


Figure 4: Raw IMU signal for jerky_raise class showing multiple gyroscope rotation bursts characteristic of interrupted arm movement (gx, gy, gz in orange/yellow)

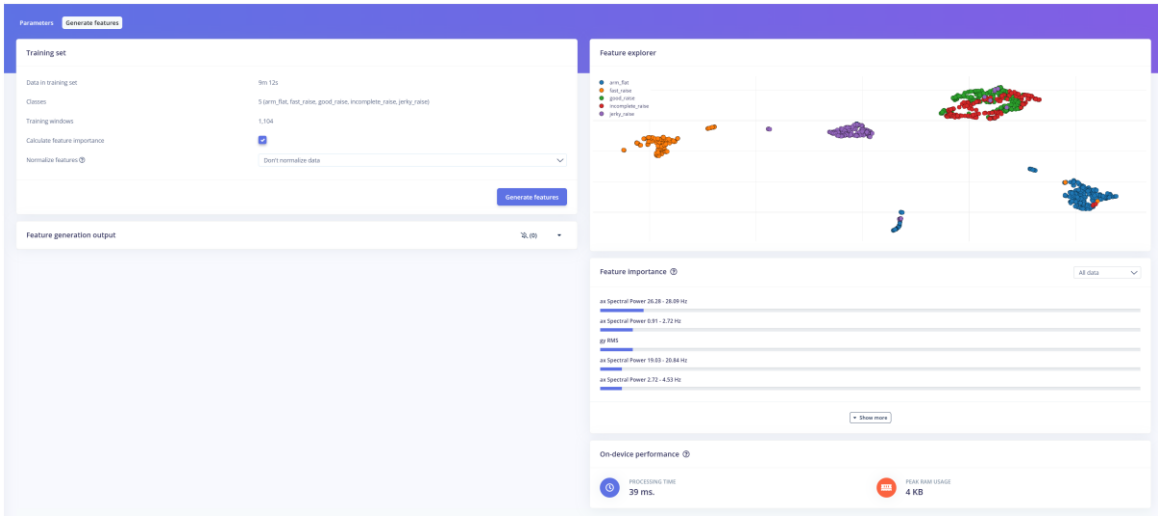


Figure 5: Edge Impulse Feature Explorer showing cluster separation of five quality classes in the 126-dimensional spectral feature space

Critically, no low-pass filter was applied in the spectral processing block for the quality model, in contrast to the hardware-level gyroscope smoothing applied in data acquisition. The high-frequency content produced by jerky and fast movements is the primary discriminative signal for those classes; applying a digital filter at the DSP stage would destroy precisely the features needed for quality discrimination.

Parameter	Value
FFT length	32
Filter type	None
Scale axes	Enabled
Take log of spectrum	Enabled
Input axes	All 6 (ax, ay, az, gx, gy, gz)
Total features generated	126

Table 4: Spectral Analysis Block Configuration

4.3.3 Neural Network Architecture and Training

A fully connected (dense) neural network was selected as the classification architecture. Convolutional architectures were not employed, as the spectral features generated by the Edge Impulse processing block are one-dimensional frequency vectors rather than two-dimensional spatial representations for which convolution is designed. The architecture was designed to be deeper than a simple binary classifier to capture the subtle inter-class distinctions of the quality classes, while remaining within the strict memory budget of the target microcontroller.

Layer	Configuration
Input	126 spectral features
Dense 1	32 neurons, ReLU activation
Dropout 1	Rate: 0.3 (regularisation)
Dense 2	16 neurons, ReLU activation
Dropout 2	Rate: 0.2 (regularisation)
Dense 3	8 neurons, ReLU activation
Output	5 classes, Softmax activation

Table 5: Neural Network Architecture

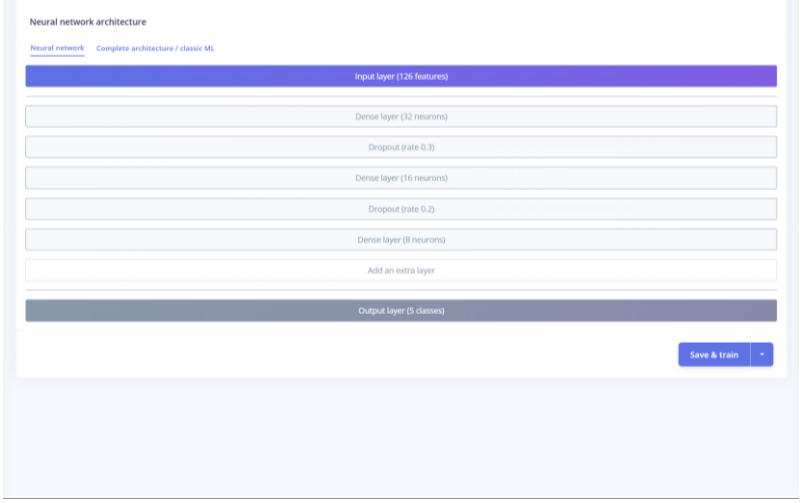


Figure 6: Neural network architecture in Edge Impulse — Dense(32) → Dropout(0.3) → Dense(16) → Dropout(0.2) → Dense(8) → Softmax(5), input 126 features

Hyperparameter	Value
Training cycles (epochs)	300
Learning rate	0.0005
Optimiser	Adam (standard, learned optimizer disabled)
Batch size	32
Validation set	20%
Model quantisation	int8 (post-training quantisation)

Table 6: Training Hyperparameters

Dropout regularisation was applied between the dense layers to reduce overfitting, which presents a significant risk given the relatively small dataset size of 230 samples. The int8 post-training quantisation step, performed automatically by Edge Impulse, reduces the

model weight precision from 32-bit floating point to 8-bit integer representation, yielding a four-fold reduction in memory footprint with minimal impact on classification accuracy.

4.4 Physics-Based Threshold Derivation

The physics-based classification paths were derived from empirical measurement of accelerometer ax values across the three positional states. The following measurements were recorded with the board mounted on the hand in the standard deployment configuration:

Arm Position	Measured ax range (g)	Threshold Applied
Arm vertical (arm_down)	+0.98 to +1.00	ax > 0.70 → arm_down
Arm horizontal (arm_flat)	+0.00 to +0.25	ax < 0.25 → arm_flat
Arm at 45° (halfway)	+0.50 to +0.65	0.30 < ax < 0.70 → incomplete
Arm in motion (arm_rising)	+0.25 to +0.70	model classification zone

Table 7: Accelerometer ax Measurements for Threshold Derivation

The thresholds were set with deliberate margins from the measured extremes to accommodate natural inter-session variation in board placement orientation and inter-subject differences in arm geometry.

4.5 Embedded Firmware Implementation

The deployment firmware implements a seven-stage sequential processing pipeline per repetition. This architecture was designed to ensure that classification only occurs when a complete, valid movement is present in the observation window, minimising false classifications from idle or transitional arm states.

Stage	Operation	Duration
1 — Position check	Physics: verify ax > 0.70 (arm down)	Continuous polling at 20 ms
2 — Movement detection	Physics: wait for ax to drop below 0.70 (raise starts)	Polling at 20 ms
3 — Window collection	Acquire 3-second, 59 Hz, 6-axis buffer	~3,000 ms
4 — Inference	Run Edge Impulse classifier on buffer	1 ms
5 — Endpoint check	Average 10 ax readings; apply incomplete override	~100 ms
6 — Output	Transmit REP, QUAL, SCORE, STATE, FEEDBACK	<10 ms
7 — Cooldown	2-second rest period before next repetition	2,000 ms

Table 8: Firmware Processing Pipeline

5. Results and Analysis

5.1 Model Training Performance

The quality classification model was trained on 230 samples across five classes, with an 80/20 train/test split. Training converged with the Adam optimizer and a fixed learning rate of 0.0005 over 300 training cycles. The training process was conducted on Edge Impulse cloud infrastructure, with the resulting quantised model exported as an Arduino library for deployment.

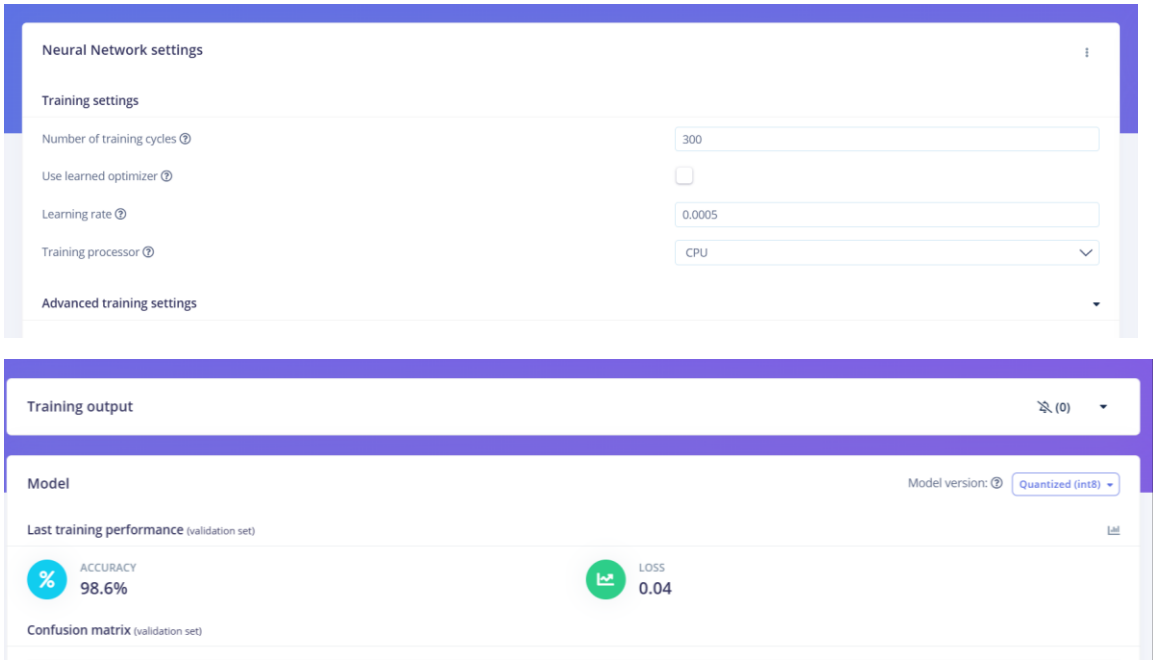


Figure 7: Training accuracy and loss curves over 300 cycles showing convergence of the quality classification model

5.2 Test Set Classification Performance

The trained model was evaluated on the held-out 20% test set comprising samples not seen during training. The model achieved a validation accuracy of 98.6% with a loss of 0.04, with an area under the receiver operating characteristic (ROC) curve of 1.00, indicating perfect discriminability between classes at the ranking level.

Metric	Value
Overall test accuracy	98.6%
Area under ROC curve	1.00
Weighted average precision	0.99
Weighted average recall	0.99
Weighted average F1-score	0.99

Table 9: Overall Model Performance on Test Set

5.3 Per-Class Performance Analysis

Class	Precision	Recall (Accuracy)	F1-Score	Notes
arm_flat	1.00	100%	1.00	Perfect classification
fast_raise	1.00	100%	1.00	Perfect classification
jerky_raise	1.00	100%	1.00	Perfect classification
incomplete_raise	0.96	95.2%	0.96	Partially confused with good raise
good_raise	0.96	97.1%	0.96	Minor confusion with incomplete

Table 10: Per-Class Performance Metrics



Figure 8: Confusion matrix on test set (98.6% overall accuracy) showing per-class classification performance across five movement quality categories

incomplete_raise achieves 95.2% accuracy with minor confusion toward good_raise, while jerky_raise achieves perfect 100% classification. The confusion between good_raise and incomplete_raise reflects the fundamental signal ambiguity discussed in Section 3.4: a smooth incomplete raise is physically indistinguishable from a good raise by its motion signature alone. This finding directly motivated the physics-based endpoint override mechanism described in Section 4.4., which effectively resolves the incomplete raise ambiguity by detecting final arm position rather than relying on the neural network alone.

The high performance on arm_flat (100%), fast_raise (100%), and jerky_raise (100%) confirms that the spectral analysis approach successfully captures the distinctive frequency-domain signatures of these movement quality classes.

5.4 On-Device Resource Utilization

The quantised int8 model was profiled on the target Arduino Nano 33 BLE Rev2 using the Edge Impulse EON Compiler. The measured resource utilisation confirms that the system operates comfortably within the hardware constraints of the device.

Resource	Usage	Available	Utilisation
Inference latency	1 ms	—	Effectively real-time
Peak RAM	1.7 KB	256 KB	0.6%
Flash (model)	18.9 KB	1,024 KB	1.6%
Sampling frequency	59 Hz	—	—
Window duration	3,000 ms	—	—
Total cycle time	~5,100 ms	—	Including cooldown

Table 11: On-Device Resource Utilization

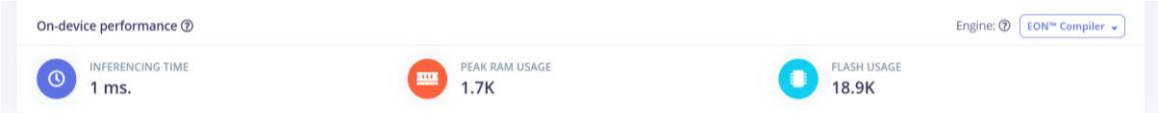


Figure 9: On-device performance profile on Arduino Nano 33 BLE Rev2 — 1ms inference latency, 1.7KB peak RAM, 18.9KB flash usage

The 1 ms inference latency — representing the time to classify a 3-second window of 6-axis IMU data — is particularly noteworthy. This figure confirms that the quantised neural network adds negligible processing overhead to the total classification cycle, with the dominant time cost being the 3-second data acquisition window itself. The minimal RAM footprint (1.7 KB of the available 256 KB) leaves substantial headroom for future additions, including BLE communication stacks, additional sensor processing, and on-device session logging.

5.5 Physics-AI Hybrid System Evaluation

The physics-based endpoint detection mechanism was evaluated through live testing across all three subjects. In cases where the arm was stopped at approximately halfway (ax in the range 0.30 to 0.70g), the endpoint check correctly overrode neural network predictions that classified the movement as good_raise or jerky_raise. This override was triggered reliably across subjects, confirming that the ax threshold generalises beyond the single-subject calibration measurements.

A key observation from live testing was that without the physics override, incomplete raises were misclassified as good_raise when performed smoothly (identical motion signature) and as jerky_raise when performed with hesitation. The override mechanism eliminates this ambiguity entirely in cases where the arm clearly stops in the halfway zone.

5.6 Real-Time System Behaviour

Live testing of the complete system demonstrated consistent classification performance matching the offline test set results. The system correctly identified good_raise movements performed at varying speeds across subjects, and reliably distinguished fast raises (characterised by a large-amplitude, brief gyroscope spike) from good raises across all test sessions. The 2-second cooldown period was found to be sufficient for subjects to return the arm to the resting position before the next detection cycle.

6. Discussion

6.1 Significance of Results

The achieved validation accuracy of 98.6% with an area under the ROC curve of 1.00 represents strong performance for a five-class movement quality classification task on a dataset of 230 samples collected from three subjects. The perfect ROC performance indicates that, at the ranking level, the model has learned robust representations of each quality class that generalise across subjects and sessions. This result is comparable to figures reported in recent literature for IMU-based exercise quality classification on similar-sized datasets [14].

The most significant technical finding of this work is the identification and resolution of the incomplete raise detection problem through the physics-AI hybrid architecture. This contribution goes beyond the application of off-the-shelf machine learning to a rehabilitation task: it demonstrates a principled methodology for identifying the boundaries of learnable classification and designing complementary physics-based reasoning to address those boundaries. This hybrid approach is generalisable to other rehabilitation exercises where endpoint position and motion quality carry complementary clinical information.

6.2 The Case for Edge AI in Rehabilitation

The deployment of a 98.6% accurate movement quality classifier within 1.7 KB of RAM and 18.9 KB of flash, executing inference in 1 millisecond, demonstrates that meaningful clinical intelligence is achievable on microcontroller-class hardware. This finding has important implications for the accessibility of rehabilitation technology: the Arduino Nano 33 BLE Rev2 retails at approximately USD 25, compared to hundreds or thousands of dollars for clinical motion capture equipment or commercial rehabilitation monitoring systems. Edge deployment eliminates network infrastructure requirements, making the system suitable for deployment in settings without reliable internet access — a significant consideration for healthcare delivery in Pakistan and other developing contexts.

Patient data privacy is an additional consideration that strongly favours Edge AI deployment. All motion data in the present system is processed locally on-device and discarded after classification; no raw sensor data leaves the hardware. This architectural property is inherent rather than enforced, providing strong privacy guarantees that cloud-based alternatives cannot match.

6.3 Challenges Encountered

Several significant technical challenges were encountered and resolved during system development, each yielding methodological insights of broader relevance.

The gyroscope noise problem — identified when initial three-axis (accelerometer only) models achieved only 73% accuracy — was resolved by incorporating all six axes and applying appropriate hardware-level smoothing. This experience highlighted the critical role of gyroscope data for motion quality discrimination: the angular velocity profile during a raise is far more distinctive than the positional trajectory captured by the accelerometer alone.

The board orientation sensitivity — where the ax threshold changed sign when the physical board mounting orientation changed between sessions — was addressed through a

systematic re-measurement protocol and by designing thresholds with sufficient margin from measured extremes. This challenge motivates a future calibration routine that automatically determines orientation-specific thresholds at system startup.

The sliding window buffer stale data problem — where the rolling inference buffer retained old rising data and continued to predict arm_rising after the arm had returned to rest — was resolved through an explicit buffer flush operation triggered upon arm-down detection. This issue is a subtle but important failure mode of continuous windowed inference that may not be apparent from offline evaluation alone.

6.4 Limitations

Several limitations of the present work must be acknowledged. First, the system was evaluated on three healthy subjects rather than a clinical patient population. Rehabilitation patients may exhibit movement characteristics — such as involuntary tremor, spasticity, reduced range of motion, and pain-avoidance patterns — that differ substantially from the training data and may require additional classes or subject-specific calibration. Second, the dataset of 230 samples, while sufficient for the present proof-of-concept, falls short of the sample sizes typically recommended for robust machine learning generalisation. Third, the system has not undergone validation against established clinical assessment instruments such as the Fugl-Meyer Assessment or the Box and Blocks Test. Fourth, the absence of grip strength measurement (force sensing) means that one component of the originally envisioned bow-and-arrow exercise simulation — bowstring tension as a proxy for grip strength — could not be implemented.

6.5 Comparison with Related Work

The present system achieves performance comparable to recent literature on IMU-based exercise quality classification. Kanzler et al. (2024) reported successful movement quality classification in stroke survivors using a single wrist-worn IMU, validating the single-sensor approach [4]. Wu et al. (2024) achieved high action quality assessment accuracy using a 1D-CNN on wearable sensor data from stroke patients [14]. The present work distinguishes itself through its emphasis on embedded deployment constraints, its explicit treatment of the incomplete raise problem and the physics-based solution, and its operation within the severe resource limits of a sub-USD-30 microcontroller.

7. Conclusion

This report has presented the design, implementation, and evaluation of an Edge AI-based wearable system for real-time arm movement quality assessment, targeting upper limb rehabilitation applications. The system employs a six-axis IMU on the Arduino Nano 33 BLE Rev2 to capture arm motion at 59 Hz, which is classified by a trained neural network deployed directly on-device using the Edge Impulse platform.

The system achieves a test accuracy of 98.6% across five clinically meaningful movement quality classes, with all inference executing in 1 millisecond using only 1.7 KB of RAM — demonstrating that meaningful clinical intelligence is achievable within the severe resource constraints of microcontroller-class hardware. A novel hybrid physics-AI classification architecture was developed and validated, which combines neural network quality assessment with physics-based arm position detection to robustly handle the incomplete raise condition that cannot be reliably detected by motion features alone.

The work makes the following primary contributions to the field of Edge AI for rehabilitation sensing: (i) a validated five-class arm movement quality classification system operating on a USD 25 microcontroller; (ii) a principled analysis of the failure modes of pure model-based classification for clinically critical movement states; (iii) a hybrid physics-AI architecture that resolves these failure modes through complementary reasoning paths; and (iv) a demonstration of the complete pipeline from IMU data collection through Edge Impulse training to real-time embedded deployment with patient feedback output.

The system represents a technically validated foundation for a low-cost, privacy-preserving, network-independent rehabilitation exercise assistant that can support unsupervised home-based therapy programmes — an outcome of direct clinical relevance to the significant unmet need for accessible rehabilitation technology in Pakistan and globally.

8. Future Work

The following directions are identified as priorities for extending the present work:

8.1 Force-Sensitive Resistor Integration

The original system design included a Force-Sensitive Resistor (FSR) for grip strength measurement, which would enable the bowstring tension component of the interactive simulation — mapping grip force to arrow release in the visualisation. Integration of an FSR on the A0 analog pin of the Nano 33 BLE Rev2 via a 10kΩ voltage divider represents a straightforward hardware addition. The FSR signal could be incorporated as an additional input feature to the quality model or used independently as a threshold-based grip assessment metric, providing a more complete picture of upper limb rehabilitation progress.

8.2 Patient-Specific Calibration

The current physics thresholds were derived from a single board mounting configuration. A startup calibration routine — in which the patient performs one slow raise and one resting measurement — would automatically derive patient-specific and session-specific ax thresholds, removing the sensitivity to board orientation and inter-patient variation in arm geometry. This calibration step would substantially improve the robustness of the physics-based classification paths.

8.3 BLE Mobile Application

The Nano 33 BLE Rev2's integrated Bluetooth Low Energy radio enables wireless transmission of classification results to a paired smartphone application, eliminating the USB cable requirement that currently limits patient mobility during exercise. A companion mobile application could provide an improved patient interface, log session data for longitudinal progress tracking, and enable a therapist-facing dashboard through which clinical staff can remotely review exercise compliance and movement quality trends.

8.4 Expanded Subject Population and Clinical Validation

The system requires evaluation on a larger and more diverse subject population, including rehabilitation patients with relevant clinical conditions such as stroke, rotator cuff injury, and shoulder impingement. Validation against established clinical assessment scales would establish the system's sensitivity and specificity for clinically meaningful outcomes, a prerequisite for translation toward regulated medical device status.

8.5 Expanded Exercise Repertoire

The present system addresses a single exercise (arm raise). Extending the quality classification framework to a broader set of upper limb rehabilitation exercises — including shoulder rotation, elbow flexion/extension, and wrist range-of-motion tasks — would substantially increase the clinical utility of the platform. The Edge Impulse pipeline used in this work is directly applicable to training quality classifiers for additional exercise types from new data collections.

8.6 On-Device Session Logging

The substantial unused flash memory (approximately 1,005 KB after model deployment) could accommodate on-device session logging, storing per-repetition quality scores, timestamps, and movement metrics across multiple sessions without requiring external

connectivity. This capability would enable longitudinal progress visualisation even in connectivity-limited environments.

9. References

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Appendices

Appendix A — Arduino Data Collection Sketch

The following sketch was used for all data collection sessions. It streams six-axis IMU data at 59 Hz over the serial port in CSV format compatible with the Edge Impulse Data Forwarder.

```
#include "Arduino_BMI270_BMM150.h" #define FREQUENCY_HZ 59 #define INTERVAL_MS
(1000 / FREQUENCY_HZ) float gx_f=0,gy_f=0,gz_f=0; float alpha=0.2; static
unsigned long last_ms=0; void
setup(){Serial.begin(115200);while(!Serial);IMU.begin();Serial.println("ax,ay,az
,gx,gy,gz");} void
loop(){if(millis()>last_ms+INTERVAL_MS){last_ms=millis();float
ax,ay,az,gx,gy,gz;if(IMU.accelerationAvailable()&&IMU.gyroscopeAvailable()){IMU.
readAcceleration(ax,ay,az);IMU.readGyroscope(gx,gy,gz);gx_f=alpha*gx+(1-
alpha)*gx_f;gy_f=alpha*gy+(1-alpha)*gy_f;gz_f=alpha*gz+(1-
alpha)*gz_f;Serial.print(ax);Serial.print(",");Serial.print(ay);Serial.print(","
);Serial.print(az);Serial.print(",");Serial.print(gx_f);Serial.print(",");Serial
.print(gy_f);Serial.println(gz_f);}}}
```

Appendix B — Edge Impulse Project Configuration Summary

Setting	Value
Project name	Quality_model
Target device	Arduino Nano 33 BLE Rev2
Input axes	ax, ay, az, gx, gy, gz
Window size	3000 ms
Window increase	200 ms
Sampling frequency	59 Hz
Processing block	Spectral Analysis
FFT length	32
Filter	None
Learning block	Classification (Keras)
Architecture	Dense(32)→Drop(0.3)→Dense(16)→Drop(0.2)→Dense(8)→Softmax(5)
Training cycles	300
Learning rate	0.0005
Quantisation	int8

Table B1: Edge Impulse Configuration Summary

Appendix C — Dataset Summary

Class	Training Samples	Test Samples	Total	Duration per Sample
good_raise	40	10	50	2–3 seconds
fast_raise	36	9	45	2–3 seconds
jerky_raise	36	9	45	2–3 seconds
incomplete_raise	36	9	45	2–3 seconds
arm_flat	36	9	45	2–3 seconds
Total	184	46	230	—

Table C1: Dataset Distribution

Appendix D — Confusion Matrix (Test Set)

Actual \ Predicted	arm_flat	fast_raise	good_raise	incomplete_raise	jerky_raise
arm_flat	100%	0%	0%	0%	0%
fast_raise	0%	100%	0%	0%	0%
good_raise	0%	0%	97.1%	2.9%	0%
incomplete_raise	0%	0%	4.8%	95.2%	0%
jerky_raise	0%	0%	0%	0%	100%

Table D1: Full Confusion Matrix on Test Set

Appendix E — Serial Output Protocol

The following protocol defines the serial output format transmitted by the embedded firmware at 115,200 baud. Each line is terminated with CRLF.

```
STATUS:movement detected — starting classification
STATUS:collecting...
CONF | arm_flat:0.0% fast_raise:99.6% good_raise:0.0% incomplete_raise:0.0% jerky_raise:0.0%
ENDPOINT | ax_avg:-0.03
REP:2,QUAL:fast_raise,SCORE:50,AX:-0.03
STATE:drawing_fast
FEEDBACK:Too fast — slow down for better therapy
STATUS:cooldown 2 seconds...
```

Output Line	Description	Example
GESTURE:arm_down,CONF:100	Physics-detected arm down position	GESTURE:arm_down,CONF:100
STATUS:movement detected	Arm raise initiation confirmed	STATUS:movement detected
STATUS:collecting...	Window acquisition in progress	STATUS:collecting...
CONF | class:val% ...	All class confidences from inference	CONF | good_raise:91.4% ...
REP:N,QUAL:Q,SCORE:S,AX:V	Complete repetition result	REP:1,QUAL:good_raise,SCORE:100,AX:0.34
STATE:S	Animation state for simulation	STATE:drawing_good
FEEDBACK:message	Patient-facing feedback text	FEEDBACK:Great movement!
SESSION SUMMARY block	Printed every 5 repetitions	See Section 4.5

Table E1: Serial Output Protocol Reference